

Multi-Satellite Coordinated Beam Hopping for Interference Mitigation Under Tilted Beam Effects: A Graph-Theoretic Approach

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Abstract—This letter presents a low-complexity multi-satellite coordinated beam hopping (BH) algorithm based on graph theory to achieve efficient resource allocation and interference mitigation (IM) in large-scale low earth orbit (LEO) satellite communication systems. We first reveal that the severe interference impact of tilted beams necessitates the joint optimization of satellite-cell association (SCA) and BH slot assignment (BHSA). To address the coupling between SCA and BHSA, we innovatively reformulate this problem as a specialized dynamic graph coloring problem, where both the graph topology and the vertex coloring are optimization variables for this problem. Consequently, we propose a two-stage algorithm integrating the minimum-cost maximum flow (MCMF), tabu search (TS), and graph coloring (GC). The MCMF algorithm efficiently generates a high quality initial SCA scheme under the limitation of satellite beam resource. The TS algorithm further performs joint optimization of SCA and BHSA, where the former is explored through neighboring solution generation, and the latter is obtained through GC. The simulation results demonstrate the superiority of the proposed algorithm, which can achieve high-quality seamless coverage with low complexity in mega-constellation and dense cell scenarios.

Index Terms—Multi-satellite cooperation, beam hopping, graph theory, resource allocation, interference mitigation.

I. INTRODUCTION

THE demand for ubiquitous connectivity in the next-generation wireless networks has spurred the rapid development of satellite-terrestrial communication systems [1]. In order to accommodate the highly dynamic and uneven traffic demands of terrestrial users, beam hopping (BH) has emerged as a promising technology that can flexibly allocate resources in multiple domains and mitigate interference. Given the intrinsically NP-hard nature of BH pattern design under numerous constraints, existing studies have explored various techniques to tackle this challenge [2], [3], [4], [5], [6],

[7], [8], [9]. Various optimization-based methods have been investigated in [2], [3], and [4] to solve the multi-satellite coordinated interference avoidance problem. However, their complexity grows exponentially with the problem dimensionality. In contrast, reinforcement learning (RL) offers a scalable alternative via learning adaptive policies, yet it requires enormous datasets and computational resources for training, and has difficulty in generalizing beyond the training set [5], [6], [7]. Therefore, simpler and more stable traditional algorithms, such as heuristics and graph-based approaches, still hold significant value in practical systems [8].

Moreover, tilted satellite beams will not only continue to exist in future large-scale constellations, but also enable more flexible utilization of satellite resources [10], yet most existing works neglect their severe interference impact. [11] explicitly emphasizes that these beams have exceptionally large and irregular interference footprints. Unlike beams with large elevation angles, the interference footprint of a tilted beam is determined by the positions of both the satellite and the target cell. In such cases, assessing interference by simple criteria such as distance or beam density in [2], [5], [6], and [7] will lead to interference misjudgment, leaving the severe interference of tilted beams unmitigated in certain areas. In order to achieve effective IM, the satellite-cell association (SCA) and BH slot assignment (BHSA) should be jointly optimized rather than decomposed and optimized separately, leading to a sharp increase in the dimensionality of the optimization problem.

Inspired by the above discussions, in this letter we establish an accurate interference model of tilted beams for precise IM and further develop a graph-based heuristic multi-satellite BH algorithm with the merits of low complexity and high transferability. The main contributions and innovations of this work are summarized as follows.

- We reveal the necessity of establishing an accurate interference model for tilted beams and jointly optimizing SCA and BHSA when designing BH patterns for IM.
- We model this coupled optimization problem as a specialized dynamic graph coloring problem, which has rarely been systematically investigated.
- We exploit the interference characteristics of tilted beams and propose a two-stage algorithm that specifically leverages the minimum-cost maximum flow (MCMF), tabu search (TS), and graph coloring (GC) to solve the dynamic GC problem.
- Simulation results demonstrate that the proposed BH algorithm exhibits a significant performance advantage over existing low-complexity BH methods.

II. SYSTEM MODEL AND PROBLEM FORMULATION

A. System Model

As shown in Fig. 1, we consider the downlink scenario of a multi-satellite beam hopping system where S satellites provide

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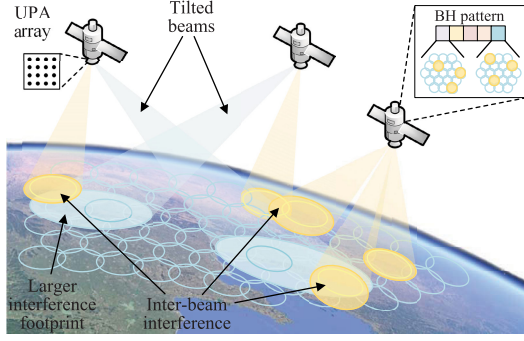


Fig. 1. Multi-satellite coordinated BH scheduling under the interference impact of tilted beams.

coverage for C ground cells in service region $\mathcal{A}$ with high service requirements. The set of satellites and cells are denoted as $\mathcal{S} = \{1, 2, \dots, S\}$ and $\mathcal{C} = \{1, 2, \dots, C\}$, respectively. Due to hardware limitations and heterogeneous service loads among satellites, we assume that satellite s can generate at most L_s spotbeams simultaneously to serve the region $\mathcal{A}$. The period of BH is T_p , which contains T BH slots that form the set $\mathcal{T} = \{1, 2, \dots, T\}$. Hence, the BH pattern at slot $t \in \mathcal{T}$ can be represented as

$$\mathbf{x}_t = [x_{t,1}, x_{t,2}, \dots, x_{t,C}]^T \in \mathbb{N}^{C \times 1}, \quad (1)$$

where $x_{t,c}$ signifies the index of the satellite serving cell c at BH slot t . In particular, we let $x_{t,c} = 0$ if the corresponding cell is not served at this BH slot. For the whole BH period, the beam pattern matrix is denoted as $\mathbf{X} = [\mathbf{x}_1, \mathbf{x}_2, \dots, \mathbf{x}_T] \in \mathbb{N}^{C \times T}$. In this letter, we assume that each cell is served by one satellite in a single BH slot, which can ensure seamless coverage with the minimal resource usage. Hence, we denote the serving satellite of cell c as s_c and the corresponding serving BH slot as t_c , both of which are uniquely determined by $\mathbf{X}$. The satellite-cell association scheme and BH slot assignment scheme are further defined as $\mathbf{s} = [s_1, s_2, \dots, s_C] \in \mathbb{N}^{1 \times C}$ and $\mathbf{t} = [t_1, t_2, \dots, t_C] \in \mathbb{N}^{1 \times C}$, respectively. In order to improve spectrum efficiency, we assume that all spotbeams occupy the entire downlink bandwidth B .

Each satellite is equipped with a uniform planar array (UPA) composed of $N_x \times N_y$ antennas, and the DFT beamforming in [11] is employed to generate spotbeams. At the user side, we assume that the receive antenna pattern is omnidirectional with $G_r = 1$. Therefore, denote the cell to which location p belongs as c_p , the signal-to-interference-plus-noise ratio (SINR) at location p is calculated as

$$\text{SINR}_p = \frac{P_b |h(s_{c_p}, c_p, p)|^2}{\sigma^2 + \sum_{c \in \mathcal{C} \setminus \{c_p\}} P_b |h(s_c, c, p)|^2 \delta(t_{c_p} - t_c)}, \quad (2)$$

where $|h(s_{c_p}, c_p, p)|^2$ is the channel gain, P_b is the power of the satellite spotbeams, $\delta(x) = 1$ if $x = 0$ and 0 otherwise. $\sigma^2 = \kappa T_n B$ represents the noise power of receivers, where κ denotes the Boltzmann's constant and T_n the noise temperature of receivers, respectively.

B. Problem Formulation

Unlike most existing works which assess interference simply based on spatial distance between cells or the number of active beams in the same BH slot [2], [5], [6], [7], in this letter we adopt a more accurate interference model by taking

the interference impact of tilted beams into account. When the elevation angle of a spotbeam decreases, the interference footprint becomes distorted and stretched, and the interference area increases sharply [11], as illustrated in Fig. 1. Hence, when a cell is served by different satellites, the set of interfered cells also varies, which necessitates the joint optimization of SCA and BHSA to achieve effective IM. For the purpose of capturing the interference of tilted beams more accurately while preserving the robustness of IM for uninterrupted service, we define the following interference indicator

$$J(s, c, i) = \begin{cases} 1, & \max_{\{p|c_p=i\}} P_b |h(s, c, p)|^2 > I_{\text{thr}} \\ 0, & \text{otherwise} \end{cases}, \quad (3)$$

with I_{thr} denoting the interference threshold of the system. We let $J(s, c, i) = 1$ for strong interference greater than I_{thr} , indicating that if cell c is served by satellite s , cell i cannot be scheduled in the same BH slot, otherwise we let $J(s, c, i) = 0$.

The goal of multi-satellite BH scheduling is to maximize the minimum SINR within the service region under time and satellite beam resource constraints, thereby ensuring service availability for users even in areas with strong interference. Hence, the optimization problem can be formulated as follows:

$$\begin{aligned} P_1 : & \max_{\mathbf{s}, \mathbf{t}} \min_{p \in \mathcal{A}} \text{SINR}_p \\ \text{s.t. } & C_1 : \sum_{t=1}^T (1 - \delta(x_{t,c})) = 1, \forall c \in \mathcal{C}, \\ & C_2 : \sum_{c=1}^C \delta(s - x_{t,c}) \leq L_s, \forall s \in \mathcal{S}, \forall t \in \mathcal{T}. \end{aligned} \quad (4)$$

C_1 indicates that each cell is served by one satellite in a single BH slot. C_2 limits the maximum number of simultaneously active spotbeams per satellite. Problem P_1 is an intractable NP-hard nonlinear integer programming problem. Due to the intrinsic correspondence between IM and graph coloring (GC), both of which impose constraints that conflicting entities cannot share the same resource [12], we build a constraint

$$C_3 : J(s_c, c, i) x_{t,c} x_{t,i} (c - i) = 0, \forall c, i \in \mathcal{C}, \forall s \in \mathcal{S}, \forall t \in \mathcal{T}, \quad (5)$$

and transform P_1 into a GC-based problem

$$P_2 : \min_{\mathbf{s}, \mathbf{t}} I_{\text{thr}} \quad \text{s.t. } C_1, C_2, C_3. \quad (6)$$

C_3 guarantees that if there is interference between the spotbeams serving cells i and c , they are not allowed to be illuminated simultaneously. By identifying the minimum I_{thr} under which the GC problem remains feasible, harmful interference is eliminated to the utmost degree and the strongest residual interference is controlled below I_{thr} , thus improving the minimum SINR within $\mathcal{A}$.

Unlike conventional GC problems with static graphs, the graph topology in P_2 is determined by the SCA $\mathbf{s}$ since different satellite selections for a cell result in distinct spotbeam interference footprints, thereby altering the edges connected to that cell's vertex, as illustrated in Fig. 2. Therefore, P_2 is a dynamic GC problem that involves the joint optimization of SCA $\mathbf{s}$ and BHSA $\mathbf{t}$. After $\mathbf{s}$ is determined, the interference graph $G_c(\mathbf{s}) = (V_c, E_c(\mathbf{s}))$ can be constructed by the vertex set $V_c = \mathcal{C}$ and the edge set $E_c(\mathbf{s}) = \{(u, v) \mid J(s_u, u, v) = 1\}$, and $\mathbf{t}$ can be obtained by solving the GC problem on $G_c(\mathbf{s})$.

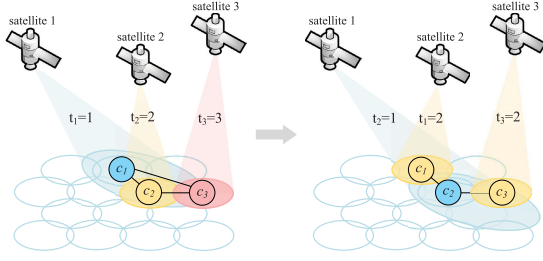


Fig. 2. An example of the dynamic GC problem. If $\mathbf{s} = [1, 2, 3]$, 3 BH slots are required for IM. If $\mathbf{s}$ is optimized to $[2, 1, 3]$, $E_c(\mathbf{s})$ changes accordingly and IM can be achieved with only 2 BH slots.

III. GRAPH-BASED MULTI-SATELLITE BEAM HOPPING SCHEDULING WITH INTERFERENCE MITIGATION

In this section, we elaborate a two-stage algorithm that specifically leverages the minimum-cost maximum flow, tabu search, and graph coloring to solve P_2 with low complexity. In stage 1, we utilize the MCMF algorithm to acquire a high-quality initial solution to $\mathbf{s}$, which significantly accelerates the convergence of the subsequent optimization. In stage 2, $\mathbf{s}$ and $\mathbf{t}$ are further optimized jointly via TS and GC to obtain the final solution to P_2 .

A. MCMF-Based Initial SCA

Selecting satellites with a large elevation angle for cells in $\mathbf{s}$ can significantly reduce the interference area of spotbeams, thus increasing the probability of obtaining high-quality solutions of $\mathbf{t}$. Hence, maximizing the sum of elevation angles for all beams in the system serves as an effective proxy for a low-interference SCA scheme. However, simply employing the maximum elevation angle principle would result in some satellites serving numerous cells simultaneously, which leads to the violation of C_2 . To address this issue, we introduce the MCMF algorithm to obtain an initial SCA scheme that maximizes the elevation angles while satisfying C_2 .

The goal of the MCMF problem is to find a maximum flow in a directed network $G_f = (V_f, E_f)$ with capacity $c(u, v) \geq 0$ and cost $w(u, v)$ per unit flow on each edge, such that the total cost $w(f) = \sum_{(u,v) \in E_f} w(u, v) \cdot f(u, v)$ is minimized, where $f(u, v)$ is the flow of the edge (u, v) [13]. In our scenario, G_f can be constructed by transforming C_2 and elevation angles into the capacity and cost of the edges, respectively. After introducing a source node v_S and a sink node v_T , the set of vertices V_f can be written as

$$V_f = \{v_S\} \cup \{v_T\} \cup \mathcal{S} \cup \mathcal{C}, \quad (7)$$

and the edges in set E_f are constructed by the following rule:

$$[c(u, v), w(u, v)] = \begin{cases} [L_v T - \Delta_L, 0], & u = v_S, \forall v \in \mathcal{S}, \\ [1, -\theta_e^{u,v}], & \forall u \in \mathcal{S}, \forall v \in \mathcal{C}, \\ [1, 0], & \forall u \in \mathcal{C}, v = v_T, \\ [0, 0] \text{ (no edge)}, & \text{otherwise,} \end{cases} \quad (8)$$

where $\theta_e^{u,v}$ is the elevation angle of the satellite u at the center of the cell v . Fig. 3 explains the construction of G_f . It should be noted that edges with capacity of 1 regulate that each cell is only served by one satellite, and edges with capacity of $L_v T - \Delta_L$ aim at satisfying C_2 . We introduce the non-negative integer margin Δ_L to improve robustness,

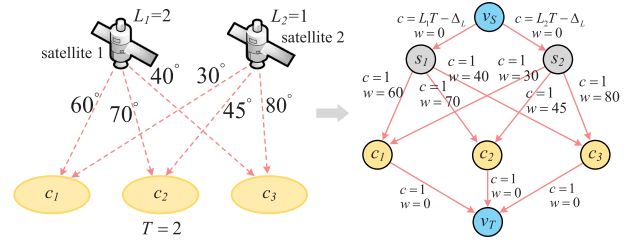


Fig. 3. The construction of the cost-augmented flow network.

ensuring that C_2 is always satisfied for the BH pattern $\mathbf{X}$ after optimizing BHSA $\mathbf{t}$ in the next stage. After solving the MCMF problem, $\mathbf{s}$ with the largest sum elevation angle of the system $\sum_{c=1}^C \theta_e^{s_c, c}$ under constraint C_2 can be obtained, which serves as an initial solution for subsequent optimization.

In this letter, we employ the Cost-Augmented Edmonds-Karp algorithm to solve the MCMF problem [13], whose time complexity is $\mathcal{O}(F_m(|E_f| + |V_f| \log |V_f|))$, where F_m is the maximum flow of G_f , $|\cdot|$ denotes the cardinality of a set. Since $|V_f| = S + C$ and $|E_f| = S + C + SC$, the complexity can be rewritten as $\mathcal{O}(F_m SC)$.

B. Joint Optimization of SCA and BHSA by TS and GC

After obtaining a high-quality initial $\mathbf{s}$ by MCMF, we further leverage TS and GC to optimize $\mathbf{s}$ and $\mathbf{t}$ jointly. TS is a metaheuristic local search algorithm that iteratively navigates through the solution space and avoids cyclic search using a tabu list [12], [14]. During each iteration of TS, we first explore different $\mathbf{s}$ by generating N_n neighboring SCA schemes $\mathbf{S}' = (\mathbf{s}^{(1)}, \mathbf{s}^{(2)}, \dots, \mathbf{s}^{(N_n)})$. Specifically, for the interference graph $G_c(\mathbf{s}^*)$ corresponding to the current optimal SCA scheme $\mathbf{s}^*$, we first record the end vertices of all conflicting edges (edges whose end vertices are assigned the same color) in $G_c(\mathbf{s}^*)$, which are denoted as set $V_{\text{con}}(\mathbf{s}^*)$. Then, each neighboring solution is generated as follows:

$$s_c^{(i)} = \begin{cases} s_r, & c = c_r, \\ s_c^*, & \text{otherwise,} \end{cases} \quad \forall c \in \mathcal{C}, \quad (9)$$

where c_r is a random cell in $V_{\text{con}}(\mathbf{s}^*)$ and s_r is a random satellite satisfying $J(s_r, c_r, v_{\text{con}}) = 0$, in which v_{con} is the other end vertex of the conflicting edge (c_r, v_{con}) on $G_c(\mathbf{s}^*)$. This strategy ensures that at least one conflicting edge on $G_c(\mathbf{s}^*)$ is eliminated for each neighboring solution, thus mitigating the time-domain-intractable interference through the space domain. Once a neighboring solution $\mathbf{s}^{(i)}$ is specified, the corresponding interference graph $G_c(\mathbf{s}^{(i)})$ is consequently determined, on which the BHSA $\mathbf{t}^{(i)}$ can be obtained by solving the GC problem. Since T colors may not suffice to achieve a conflict-free coloring on $G_c(\mathbf{s}^{(i)})$, we minimize the number of conflicting edges $f_{\text{con}}(G_c(\mathbf{s}^{(i)}), \mathbf{t})$ instead. This problem can be efficiently solved by the HEAD algorithm [15], which is one of the most powerful GC algorithms demonstrated in [16] and [17]. After obtaining the BHSA $\mathbf{t}^{(i)} = \arg \min_{\{\mathbf{t} \in \mathcal{F}_t(\mathbf{s}^{(i)})\}} f_{\text{con}}(G_c(\mathbf{s}^{(i)}), \mathbf{t})$ by HEAD, where $\mathcal{F}_t(\mathbf{s}^{(i)})$ denotes the feasible set of $\mathbf{t}$ such that the pair $(\mathbf{s}^{(i)}, \mathbf{t})$ satisfies C_2 , we use the minimum number of conflicting edges $f_{\text{con}}(G_c(\mathbf{s}^{(i)}), \mathbf{t}^{(i)})$ to evaluate the quality of $\mathbf{s}^{(i)}$. Moreover, in order to avoid revisiting previously explored solutions, a tabu list $L = \{\mathbf{s}_{-1}^*, \mathbf{s}_{-2}^*, \dots, \mathbf{s}_{-l_{\text{tabu}}}^*\}$ records the solutions of the last l_{tabu} steps at each iteration, where

Algorithm 1 MCMF-TS-GC Algorithm

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1: Input:  $\{J(s, c, i)\}_{s \in \mathcal{S}, c, i \in \mathcal{C}}, \{L_s\}_{s \in \mathcal{S}}, N_n, N_{it}, I_{thr}^-, \Delta I$ .
2: Stage 1: MCMF-based initial SCA.
3: Build  $G_f = (V_f, E_f)$  by equation (7) and (8).
4: Solve the MCMF problem on  $G_f$  by Cost-Augmented Edmonds-Karp algorithm and obtain initial solution to  $\mathbf{s}^*$ .
5: Stage 2: Joint optimization of SCA and BHSA based on TS and GC.
6: Initialize  $L$  as an empty tabu list,  $i = 0$ ,  $I_{thr} = I_{thr}^-$ .
7: repeat
8:   if  $i = N_{it}$  do
9:      $i = 0$ .
10:     $I_{thr} = I_{thr} + \Delta I$ .
11:   end if
12:   Generate  $\mathbf{S}' = (\mathbf{s}^{(1)}, \mathbf{s}^{(2)}, \dots, \mathbf{s}^{(N_n)})$  according to  $\mathbf{s}^*$ .
13:   Construct  $G_c(\mathbf{s}^{(1)}), G_c(\mathbf{s}^{(2)}), \dots, G_c(\mathbf{s}^{(N_n)})$ .
14:   Get  $f_{con}(G_c(\mathbf{s}^{(1)}), \mathbf{t}^{(1)}), f_{con}(G_c(\mathbf{s}^{(2)}), \mathbf{t}^{(2)}), \dots, f_{con}(G_c(\mathbf{s}^{(N_n)}), \mathbf{t}^{(N_n)})$  by HEAD algorithm.
15:    $\mathbf{i}^* = \arg \min_{i \in \{1, 2, \dots, N_n\}, \mathbf{s}^{(i)} \notin L} f_{con}(G_c(\mathbf{s}^{(i)}), \mathbf{t}^{(i)})$ .
16:    $\mathbf{s}^* = \mathbf{s}^{(\mathbf{i}^*)}$ ,  $\mathbf{t}^* = \mathbf{t}^{(\mathbf{i}^*)}$ .
17:    $L = \{\mathbf{s}^*, \mathbf{s}_{-1}^*, \dots, \mathbf{s}_{-l_{tabu}+1}^*\}$ .
18:    $i = i + 1$ .
19: until  $f_{con}(G_c(\mathbf{s}^*), \mathbf{t}^*) = 0$ 
20: Map  $\mathbf{s}^*$  and  $\mathbf{t}^*$  back to BH pattern matrix  $\mathbf{X}$ .
21: Output:  $\mathbf{X}$ .

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$\mathbf{s}_{-i}^*$ represents the best solution acquired i iterations earlier. As a result, at the end of each iteration, we find the index of the best neighboring solution

$$\mathbf{i}^* = \arg \min_{i \in \{1, 2, \dots, N_n\}, \mathbf{s}^{(i)} \notin L} f_{con}(G_c(\mathbf{s}^{(i)}), \mathbf{t}^{(i)}), \quad (10)$$

and update $\mathbf{s}^*, \mathbf{t}^*, L$ as

$$\mathbf{s}^* = \mathbf{s}^{(\mathbf{i}^*)}, \mathbf{t}^* = \mathbf{t}^{(\mathbf{i}^*)}, L = \{\mathbf{s}^*, \mathbf{s}_{-1}^*, \dots, \mathbf{s}_{-l_{tabu}+1}^*\}, \quad (11)$$

respectively.

Finally, we maximize I_{thr} , which is the ultimate goal of P_2 , by incorporating the aforementioned TS-GC algorithm into a monotonic feasibility test. Specifically, we initialize I_{thr} at a conservative lower bound I_{thr}^- and gradually increase it until C_3 can be satisfied, i.e., $f_{con}(G_c(\mathbf{s}^*), \mathbf{t}^*) = 0$. When testing the feasibility of each I_{thr} , we execute the TS for N_{it} iterations. If $f_{con}(G_c(\mathbf{s}^*), \mathbf{t}^*) = 0$ is obtained, the current value of I_{thr} is regarded as the solution to P_2 and the test terminates, otherwise we increase I_{thr} by ΔI and continue testing, where ΔI is a fixed increment.

The time complexity of stage 2 can be estimated as $\mathcal{O}(N_{ft} N_{it} N_n t_n)$, where N_{ft} is the number of different I_{thr} tested and t_n is the complexity of addressing each neighboring solution. It is noted that t_n is mainly determined by the complexity of HEAD. The MCMF-TS-GC algorithm is summarized in **Algorithm 1**.

In practical systems, the proposed algorithm is deployed in an ephemeris-based offline pre-solving and distributed execution manner. The complete BH scheme over the entire ephemeris period is solved offline at each sampling time instant, while each satellite only stores and executes its

assigned BH actions during operation, thereby enabling parallel and low-complexity implementation of the proposed method in large-scale dynamic constellations.

IV. SIMULATION RESULTS AND PERFORMANCE ANALYSIS

In this section, we set up simulations on a mega LEO constellation with 10800 satellites to evaluate the performance of the proposed method. Without loss of generality, we consider a 350km-radius circular service region $\mathcal{A}$ with dense ground cells, which is covered by $S = 25$ satellites sampled from the constellation in the vicinity of $\mathcal{A}$ operating at two orbital heights $H_1 = 510\text{km}$ and $H_2 = 980\text{km}$. We simulate two representative scenarios in practical engineering:

- Case 1 (Low cell density): The cell radius is set to 30km, resulting in a total of $C = 148$ cells. The maximum number of simultaneously active beams is set as $L_s = 8$.
- Case 2 (High cell density): The cell radius is set to 12.5 km, with $C = 928$ and $L_s = 32$.

Moreover, according to the parameters of 3GPP standard [10], the downlink carrier frequency is 2GHz and the total bandwidth is $B = 30\text{MHz}$. The power of each spotbeam is set as $P_b = 50\text{W}$. A noise temperature of $T_n = 400\text{K}$ is assumed for the user terminals. Based on link budget analysis and system simulation results, we set the MCMF margin $\Delta_L = 30$, number of neighboring solutions per TS iteration $N_n = 20$, number of TS iterations $N_{it} = 10$, low interference threshold $I_{thr}^- = -125\text{dBW}$ and the increment $\Delta I = 1\text{dBW}$. To balance time resource reuse efficiency and IM ability, we select the range of 13 to 29 for T without loss of generality.

In this section, we compare the following BH schemes:

- ‘**WMIS**’: A BH pattern design method based on the weighted maximum independent set (WMIS) of the interference graph [8].
- ‘**Greedy**’: A greedy BH method proposed in [11] with the IM strategy minimizing the number of simultaneously active beams.
- ‘**NITB**’ (Neglect the interference of tilted beams): The MCMF-TS-GC algorithm is employed, but inter-beam interference is evaluated only based on spatial distance separation between cells [2], [5], [6].
- ‘**Gurobi**’: a renowned commercial solver which can provide a reference upper bound of the problem [18].
- ‘**MCMF-TS-GC**’: The proposed MCMF-TS-GC algorithm with accurate interference modeling of tilted beams.

Fig. 4 (a) and Fig. 4 (b) show the service satisfaction rate of cells and minimum SINR across the service region $\mathcal{A}$ under different BH period T , respectively. A cell is considered served if the average SINR within it exceeds 16 dB. The proposed MCMF-TS-GC BH algorithm succeeds in satisfying the service of almost all cells within a BH period of $T = 19$, and further achieves a minimum SINR of 20 dB at $T = 27$ for Case 1 and 16 dB at $T = 29$ for Case 2. Compared with the reference upper bound obtained by Gurobi, the proposed algorithm achieves near-optimal performance with dramatically reduced computational complexity. The performance gap between ‘MCMF-TS-GC’ and ‘NITB’ reflects the importance of considering the interference impact of tilted beams. The WMIS baseline achieves comparable service satisfaction rate of cells to MCMF-TS-GC when T is large, but the performance drops sharply as T decreases since it tends to equalize the SINR across most regions toward the minimum SINR. The

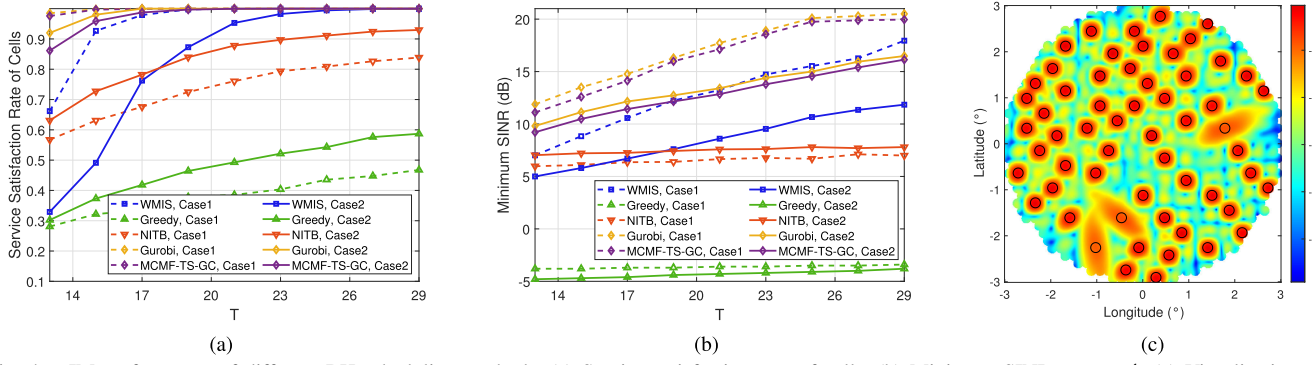


Fig. 4. IM performance of different BH scheduling methods. (a) Service satisfaction rate of cells. (b) Minimum SINR across $\mathcal{A}$. (c) Visualization of the ground receive power (in dBW) at a BH slot obtained by MCMF-TS-GC algorithm.

TABLE I
AVERAGE RUNTIME OF DIFFERENT BH METHODS

Algorithms	Runtime (s)		Complexity
	Case 1	Case 2	
WMIS	0.78	39.46	$\mathcal{O}(S^2 C^2 T^2)$
Gurobi	30.00	6000.00	Exponential [18]
MCMF-TS-GC	0.11(MCMF)	1.65(MCMF)	$\mathcal{O}(F_m SC + N_{ft} N_{it} N_n t_n)$
	1.56(TS-GC)	7.86(TS-GC)	

greedy method based on minimizing beam conflict probability shows limited performance in both cases, indicating that the number of active beams illuminated simultaneously is insufficient to capture the actual interference.

Fig. 4 (c) visualizes the beam illumination pattern for the service region at a BH slot, demonstrating the effectiveness of the proposed MCMF-TS-GC algorithm in mitigating interference for tilted beams. It can be observed that although tilted beams cause harmful interference to numerous cells, the interference footprint of all active beams do not overlap after the MCMF-TS-GC BH scheduling.

Table I provides the average runtime of different BH methods. Specifically, it can be observed that the complexity of the proposed method exhibits a much lower growth rate with the increase of C than that of WMIS, demonstrating significant advantages in dense cell scenarios. The reason is that the interference graph of WMIS contains SCT vertices instead of only C in the proposed method, leading to an extremely high time complexity of $\mathcal{O}(S^2 C^2 T^2)$.

V. CONCLUSION

This letter identified the interference impact of tilted beams and proposed a graph-based multi-satellite coordinated beam hopping algorithm to mitigate interference effectively and ensure seamless coverage in this scenario. By observing that the interference footprint of tilted beams is extensive and irregular, we formulated the interference-aware BH pattern design problem as a dynamic graph coloring problem. On this basis, we proposed an MCMF-TS-GC algorithm to solve this problem with low complexity by jointly optimizing SCA and BHSA. Simulation results validated the superiority of the proposed algorithm. In future work, we will extend the proposed BH algorithm to dynamic LEO constellations.

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